Uploading the Source Code

Where before we have used the Serial functions to send data from the Arduino board to our Serial Monitor, for the purpose of monitoring analog values, the sketch in Listing 10-1 will instead allow us to send data to the Arduino board that has been entered in the Serial Monitor. On the surface, all we need to do is type in an angle for our servo from 0 to 180, corresponding to the angle that we want to position a servo, and hit enter. When the Arduino receives this information, it will process it and move the servo to the proper angle.

We see that because serial communications happens in one-byte chunks and because of the habit of serial to send information in characters rather than numerical values (more on that shortly), we need to do a little work to get the characters that are received converted to proper numbers that can be used to position a servo. So let's upload the following code and have a look at how all this works.

Listing 10-1. *Serial Servo Source Code*

```
#include <Servo.h>

Servo servo;

void setup() {
  Serial.begin(9600);
  servo.attach( 10);
}

void loop() {
  int angle = 0;
  if (Serial.available()) {
    byte incomingByte = Serial.read();

    while (incomingByte != '\n') {
      if (incomingByte >= '0' && incomingByte <= '9')
        angle = angle * 10 + (incomingByte - '0');
      incomingByte = Serial.read();
    }

    if (angle >= 0 && angle <= 180) {
      servo.write(angle);
      Serial.println(angle);

    } else Serial.println("Choose an angle between 0 and 180.");
  }
}
```

Source Code Summary

Just like our example from the last chapter, at the beginning of our sketch we include and create a new instance for the Servo library. While it acts like any other library, we do not need to create an instance of Serial because it is already included as a part of our sketch as standard. In our `setup()` function, we have the necessary statements to establish the speed of our serial communications and attach our servo to a pin number. Our `loop()` function begins by starting our servo angle off at 0º and then checks to see if